Situation-Based Approaches in Semantics
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Handout 1
Situations, Facts, and States of Affairs, and their Roles in Semantics

1. Situations, facts, states of affairs

1.1. What are situations intuitively?

(1) a. the situation of John being angry
    b. the situation of the room being dark

'An object having a property at a time'
'Objects having a property at a time'
'Objects standing in a relation at a time'

Questions the formulations raise

- Are situations composed of objects and properties/relations, and in what way?
- Are objects and properties parts of situations?
- How specific need the properties/relations constitutive of situations be?
- Can situations be quantificational, disjunctive, or negative?

Situations need not be factual
Situations are generally considered parts of possible worlds.

1.2. Explicit reference to situations

The term 'situation' is generally used as a technical term in semantics, not restricted by constraints on the noun *situation* in natural language.

Yet our intuitions about situations to an extent can be sharpened by regarding the way natural language permits explicit reference to situations:

(2) a. the situation in which it is raining

b. the situation in which Bill kicked the dog

Constraints on explicit situation reference:

(3) a. ??? the situation in which Charlie is a horse

b. ??? the situation in which Joe resembles Bill

Same constraint on *in those circumstances*:

(4) a. It might be raining. In those circumstances, the party will be cancelled.

b. ??? The university might be in the city center. In those circumstances, Joe will probably take the job.

What we refer to as situations or circumstances must involve accidental and episodic properties. That is not how situations are understood in the semantic metalanguage.

The clausal modifier of *situation*:

(5) a. * the situation that it is going to rain

In contrast, specificational sentences are fine:

(5) b. The situation is that it is going to rain.

German:

(6) a. die Situation, in der es regnet / * dass es regnet

b. Die Situation ist, dass es regnet

c. ?? die Sachlage, dass / in der es regnet

d. Die Sachlage ist, dass es regnet.

Language-particular differences: e.g., Russian allows for (5a) (cf. Bondarenko).

Other ways in which to use the object language to sharpen our intuitions about situations:

<u>Part-related expressions applying to situations</u>

(7) a. That it was raining was part of the difficult situation in which Joe found himself.

b. ??? The car was part of the situation in which John found himself.

c. The rain was part of the difficult situation in which Joe found himself.

Existence predicates and situations

(8) a. The situation still obtains.

b. The situation does not exist.

c. The situation ??? happened / ? occurred yesterday.

1.3. Facts

Canonical fact descriptions: *the fact that* S

No semantic constraints on canonical fact descriptions

(9) a. the fact that it is raining

b. the fact that Charlie is a horse

c. the fact that Mary resembles Sue

That-clauses as fact descriptions:

(10) a. John is aware of the fact that he made a mistake.

b. John is aware that he made a mistake.

(11) a. The fact that it is raining is unfortunate.

b. That it is raining is unfortunate.

(12) a. John knows that it is raining.

b. ?? John knows the fact that it is raining.

Facts vs. situations

Situations need not be factual

Are facts as concrete as situations?

Conceptions of facts (Mulligan 'Facts' SEP article)

- Facts as instances of properties or relations, facts as truthmakers
- Facts as true propositions, as the truth of propositions, as propositions qua being true

Two historical notions of facts: Austin (1979) vs. Strawson (1971) (see also Fine 1982)

1. Austin's (1979) notion of a worldly fact

The linguistic locution appealed to: *the facts*

(13) a. The facts show that Bill is guilty.

b. That Bill is guilty is not supported by the facts.

Facts are concrete, that is, fully specific.

Facts are not quantificational or disjunctive.

2. Strawson's (1971) notion of a non-worldly fact

Linguistic locution appealed to:

Canonical fact descriptions: *the fact that* S

<u>Given canonical fact descriptions</u>

Facts can be disjunctive:

(14) the fact that either John or Mary won the race

Facts can be negative:

(15) the fact that John did not win the race

Facts can be quantificational:

(16) a. the fact that no student passed the exam

b. the fact that at least two students passed the exam

c. the fact that everyone is happy

Facts can be unspecific:

(17) the fact that the apple is red

<u>Construing facts in terms of propositions (that is, the content of sentences)</u>

Three options:

1. Facts as true propositions

But true propositions have other properties:

(18) a. John believes the true proposition that the world is round

b. ??? John believes the fact that the world is round.

(19) a. John noticed the fact that it is raining.

b. ??? John noticed the true proposition that it is raining.

(20) a. The fact that it is raining is surprising.

b. ??? The true proposition that it is raining is surprising.

2. A fact as 'the truth of a proposition', that is, an instance of truth in a proposition

Problem:

'The truth of a proposition' requires having the notion of a proposition, but not so facts.

(21) a. ? John noticed the truth of the proposition that it is raining.

b. ? The truth of the proposition that it is raining is surprising.

3. Facts as abstractions from true propositions

Facts implicitly defined in terms of their identity and existence conditions:

(22) a. The fact that S exists iff the proposition that S is true

b. The fact that S = the fact that S' iff the proposition that S = the proposition that S'

Issues about facts

The mode of being of facts: obtaining, but not time-relative

(23) a. The fact that it is raining holds / obtains / exists / ?? occurred / ?? happened.

b. ??? The fact that it is raining still obtains / goes on.

c. The situation in which it rains still obtains.

What would be the parts of facts? – not objects or properties

(24) a. The fact that all the students failed the exam is *partly* unexpected.

b. Joe is aware of *part of* the fact that the students failed the exam.

1.3. States of affairs

States of affairs need not be factual

The difference between propositions and states of affairs

1. Propositions are generally considered the contents of propositional attitudes, and as such display failure of substitutivity of co-referential expressions:

(25) a. The belief that the neighbor is guilty =/= the belief that Joe is guilty.

b. The state of affairs / situation in which the neighbor is guilty = the state of affairs / situation in which Joe is guilty.

2. Propositions may involve non-existents, but not so states of affairs:

(26) the proposition that the golden mountain does not exist

2. Propositions come with truth conditions; states of affairs with conditions of obtaining

(27) a. The state of affairs in which it is raining obtains / ??? is true.

b. The proposition that it is raining is true / ??? obtains.

Explicit reference to states of affairs

That-clauses:

(28) a. That it will rain / The state of affairs in which it will rain is possible / likely /

improbable.

b. John fears that it will rain / the state of affairs in which it will rain.

States of affairs can be disjunctive, negative, quantificational, like facts.

Facts as obtaining states, or the obtaining of states of affairs.

<u>The ontology of states of affairs</u>

States of affairs as structured complexes:

The state of affairs in which Mary is happy: <HAPPY, Mary>

Problems:

- But how does this give us conditions of obtaining?
- How does this ensure predication of being happy of Mary?
- How do this avoid having properties and objects count as parts of states of affairs (or facts)?

1.4. A special case of situation: cases (Moltmann 2021)

Reference to kinds of cases (29a, 30) and specific cases (28b):

(29) a. The case in which a student fails the exam needs to be excluded.

b. The five cases in which a student fails the exam were exceptional

(30) The case in which John might not return cannot be excluded.

The ontology of cases:

situations (as truthmakers of sentences) or kinds of situations, within a space of alternatives

<u>Two kinds of alternatives</u>

- quantifier instances (29)
- epistemic alternatives (30)

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2. The semantic roles of situations

2.1. Situations as objects of direct perception

Scenes and Other Situations (Barwise 1981)*, Situations and Attitudes* (Barwise / Perry 1983)

(31) a. John saw Mary leave.

b. John heard Mary sing.

Event semantics as an alternative?

(32) ?? John heard Mary's singing (but he did not recognize Mary doing the singing).

2.2. Situations as topic situations

Austin: the utterance any sentence involves reference to a topic situation

Purposes of topic situations

1. Topic situations restrict the understanding of the predicate:

(33) a. John helped Mary out.
 b. No, he did not.

2. Topic situations restrict quantifier domains:

(34) a. Every student passed the exam.

But different situations may be required for different quantifiers:

(34) b. Everyone was asleep and monitored by a research assistant.

3. Topic situations ensure uniqueness of definite NPs:

(35) a. The violinist was outstanding.
 b. The violinist greeted the second violinist.

Different situations needed different definite NPs

4. Situation-based accounts of E-type pronouns (donkey-pronouns)

(36) a. When a farmer owns a donkey, he beats it.
 b. For any minimal situation s in which a farmer owns a donkey, the farmer in s beats the donkey in s.

Problems for situation-based accounts:

(35) a. Every student who made a mistake corrected it.
 b. Most farmers who own a donkey beat it

Why should situations be involved in such sentences?

See also Moltmann (2006) for a critique of situation-based accounts to E-type pronouns.

Questions that arise when positing topic situations for the semantic structure of sentences

1. Where are topic situations located semantically?

'Unarticulated constituents' of entire sentences?

Implicit arguments of nouns, determiner, or quantifiers?

2. What, if any, are the constraints on the semantic presence of situations?

3. Do implicit situations require a syntactic representation, by a syntactic variable for situations?

<u>Another approach to implicit situations (Recanati)</u>

Situations are not part of the semantic structure of sentences; instead, they come about by pragmatic enrichment.

Sentences are always interpreted within a nonlinguistic context involving a situation.

Limits of the approach:

- different situations for different occurrences of NPs in the sentence
- implicit quantification over situation

5. Explicit and implicit non-nominal quantification over situation

<u>Frequency expressions</u>

(36) a. John is sometimes asleep (when Mary knocks at his door).

b. When a man meets a friend, he usually greets him.

c. When John gets up, he immediately takes a shower.

(37) a. For some situation s in which Mary knock at the door, John is asleep in s.

b. For all minimal situations s in which John gets up there is an extension s' of s such that John takes a shower in s'.

<u>*Instead*-phrases</u>

(38) Instead of John performing a dance, Mary sang a song.

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2. Situations as constitutive of propositional content

2.1. The notion of propositional content or of a proposition

Propositions: the contents of (declarative) sentences

Properties of propositions:

- Comes with truth conditions.
- Can form the content of propositional attitudes
- Can be shared among different agents

<u>Two standard views of propositions</u>

(39) John likes Mary.

1. Unstructured content

Propositional content consists in the set of worlds in which the sentence is true:
the set of worlds in which John likes Mary.

2. Structured propositions

Propositional content consists, in a simple case, in an n-place relation and n-1 arguments:
<LIKE, John, Mary>

2.2. Situation Semantics

Standard situation semantics (Barwise and Perry 1983, Kratzer 2007/21)

Roughly, propositional content of a sentence S: the set of situations in which the sentence S is true.

Truthmaker semantics (Fine)

'State': cover term for anything that may act as a truthmaker, e. g., situations, events, objects.
More common cover term: 'situation'
Simplifying:
Propositional content of a sentence S
The set of states (situations) that make the sentence S true and are completely relevant for the truth of S.

Unilateral truthmaker semantics:
The propositional content of a sentence is the set of its truthmakers.
Bilateral truthmaker semantics:
The propositional content of a sentence is a bilateral content, consisting of a set of truthmakers and a set of falsity makers of the sentence.

Standard situation semantics (Barwise and Perry, Kratzer)
Inexact truthmaking
Truthmaker semantics
Exact truthmaking

Issues for truthmaker semantics

What does it mean for a situation to be an exact truthmaker?

What are the parts of truthmakers?

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Handout 2
Properties and Tropes

1. Summary so far and additions

Five types of related entities

- Situations
- States
- States of affairs
- Propositions
- Facts

Situations

- Holding of a property of an object (at a time)
- Holding of a relation among objects

Situations are maximally specific:

- Constituted by specific properties or relations
- Involve particular individuals, at a particular time.

Mode of existence of situations

A situation exists / obtains.

Remark about states

The holding of a property of an object or of a relation among objects at a time.

(1) a. A state obtains at a time.

b. A state still holds.

Are states maximally specific?

In linguistics a distinction has been made between

1. Davidsonian states (Maienborn) or concrete states (Moltmann), e.g., bodily positions, sleeping, and

2. Kimian states (Maienborn) or abstract states (Moltmann), e.g., states of owning or owing something or residing somewhere, states of resembling

The distinction seems to match the distinction between situations and states of affairs.

States of affairs

May be constituted by quantifiers, disjunctions, negations, nonspecific properties or relations

The mode of being of states of affairs

States of affairs may obtain or not obtain.

States of affairs exist whether or not they obtain!

States of affairs and propositions

Propositions are like states of affairs, but:

- They have truth conditions rather than conditions of obtaining.
- They are 'representations' and constitute the content of propositional attitudes.
- They display opacity (failure of substitution of coreferential expressions), and thus, arguably, involve modes of presentations / conceptual representations.
- They may involve nonexistent objects.

Roles of states of affairs:

- Carriers of probability, objects of certain attitudes (fear)

Linguistic representations of states of affairs vs. propositions

Clausal gerunds ('imperfect nominals'): states of affairs

That-clauses: propositions and states of affairs, depending on the embedding predicate

(2) a. John's winning the race is improbable.

b. That John will win the race is improbable.

Facts:

- May be constituted by quantifiers, disjunctions, negations, nonspecific properties or relations
- Obtain by nature

Different conceptions of facts

- Facts = obtaining states of affairs
- Facts = true propositions
- Facts = the truth of propositions
- Facts obtained, by abstraction, from obtaining states of affairs / true propositions

What situations, states of affairs, facts, and propositions share

They are, somehow constituted by properties or relations (and objects or quantifiers)

Simple representations of situations

Barwise and Perry 'Situations and Attitudes': infons

(3) a. John likes Mary

b. <LIKE, John, Mary, 1>

(4) a. John does not like Mary

b. <LIKE, John, Mary, 0>

Representation of quantified states of affairs in Situation Semantics

Making use of parametric objects: place holder for ordinary objects

(5) a. Every man is happy.

b. <<EVERY(MAN), x>, <HAPPY, x>>

Standard representation of states of affairs and propositions as structured propositions

(6) a John likes Mary.

b. <LIKE, John, Mary>

Truth conditions for structured propositions / obtaining conditions for states of affairs

(6) c. The proposition <LIKE, John, Mary> is true (in a possible world w) iff

<John Mary> is in the denotation of LIKE (in w)

c. The state of affairs <LIKE, John, Mary> obtains (in a possible world w) iff <John Mary> is in the denotation of LIKE (in w)

(7) a. John does not like Mary.

Syncategorematic treatment of *not*:

(7) b. <NOT, <LIKE, John, Mary>> is true (in a possible world w) iff <John Mary> is not in the denotation of LIKE (in w).

Categorematic treatment of *not*:

(7) c. For a structured proposition p and a possible world w, NOT^{w}(p) = true iff p is not true in w.

d. <NOT, <LIKE, John, Mary> is true (in a possible world w) iff NOT^{w}(<LIKE, John, Mary>>) = true

(8) a. John is happy and Mary is content.

b. <AND, <HAPPY, John>, <CONTENT, Mary>> is true (in a possible world w) iff <HAPPY, John> is true in w and <CONTENT, Mary> is true in w. (syncategorematic treatment of *and*)

c. For structured propositions p and q, AND^{w}(p, q) = true iff p is true in w and q is true in w.

d. <AND, <HAPPY, John>, <CONTENT, Mary>> is true (in a possible world w) iff AND^{w}(<HAPPY, John>, <CONTENT, Mary>) = true (categorematic treatment of *and*)

(8) a. John is happy or Mary is blind.

b. <OR, <HAPPY, John>, <BLIND, Mary>> is true (in a world w) iff <HAPPY, John> is true in w or <BLIND, Mary> is true in w. (syncategorematic treatment of or)

c. For structured propositions p and q and a possible world w, OR^{w}(p, q) = true iff p is true in w or q is true in w.

d. <OR, <HAPPY, John>, <BLIND, Mary>> is true (in a world w) iff OR^{w}(<HAPPY, John>, <BLIND, Mary>) = true. (categorematic treatment of *or*)

(9) For a sentence S and the structured proposition p expressed by S, <u>the extension of</u> S = the set of possible worlds w such that p is true in w.

<u>Questions that properties and relations raise</u>

- What are properties and relations?
- How to properties and relations connect with the objects to which they apply?
- Do relations come with an order among their arguments?

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2. Properties

Properties and concepts

Properties: features of objects

Concepts: mentally graspable denotations of predicates

Properties of properties

- are universals: can be possessed by different objects
- ground similarity among objects

Restricted notions of a property

Natural properties (Armstrong)

Sparse properties (Lewis)

Properties that are causally relevant (occur in natural laws)

Properties that are indispensable for the full description of the world

Unrestricted notions of a property

Non-natural properties (Armstrong)

Abundant properties (Lewis)

Properties matching a set of possible objects

Properties denoted by natural language predicates

Properties generally considered one-place

Two-place properties etc.: relations

Explicit reference to properties

Surprising constraints on explicit property reference (Moltmann 2023):

(10) a. the property of being happy

b. ?? the property of laughing

c. ?? the property of pushing the ball

d. the property of resembling Mary

e. the property of owning a house

f. the property of having a father / a cold

<u>Properties as denotations of explicit property-referring terms</u>

- Tied to adjectives, but not eventive verbs?
- Tied to the copula verbs *be* and *have*?

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3. Tropes

<u>Examples</u>

John's happiness, Socrates' wisdom, Socrates' whiteness

<u>Other terms for tropes</u>

Particularized properties, concrete property instances, property bits

Accidents, modes

<u>The term 'trope'</u>

most commonly used in contemporary metaphysics, due to Williams (1953)

<u>Ancient and medieval terms</u>

Aristotle: accident and substantial form

Medieval metaphysics: mode

<u>Properties of tropes</u>

- Resemblance:

Two tropes are exactly similar iff they are instances of the very same property.

Exact resemblance among tropes is well-reflected in natural language:

(11) a. The softness of this pillow is the same as the softness of that pillow.

b. ??? The softness of this pillow is the softness of that pillow.

- Come with a bearer:

Depend for their existence on their bearer

Also for their identity?

(12) the blackness of the sculpture = the blackness of the clay

<u>One-category trope-based ontology (Williams 1953, Campbell, Bacon, and others)</u>

- Properties: sets of resembling tropes
- Individuals: bundles of collocated tropes

Do tropes really have a (direct) spatial location?

(13) ??? John's happiness was in Munich.

See Moltmann (to appear) for discussion.

Four-category ontology (Aristotle's 'Categories')

Tropes (accidents) – individuals (primary substances)

Properties (qualities) – kinds (secondary substances)

Two-category ontology

Reducing universals to tropes and individuals:

Properties as classes or kinds of resembling tropes

Individuals as bearers of tropes

Moltmann, *Abstract Objects and the Semantics of Natural Language*, OUP 2003)

Natural language displays a two-category ontology, in its core.

Qualities: pluralities of actual and possible tropes

Kinds of individuals: pluralities of actual and possible individuals.

Reference to tropes

(14) a. the apple's redness

b. John's happiness

c. Socrates' wisdom

d. the quality of the bread.

Tropes, like situations, are fully specific

Linguistic test for specificity

Predicates of description

(15) a. Mary described John's happiness / Socrates' wisdom.

b. ??? Mary described John's being happy / Socrates being wise.

c. ??? Mary described the fact that John is happy / that Socrates was wise

d. Mary described the situation in which John fell down.

e. Mary described John's fall.

Result of applying the test

Tropes, events, and situations are fully specific.

States of affairs, facts are not.

Note that the test does not distinguish types of objects, but rather descriptions, namely as to whether they fully display the nature of the object or not:

(16) Mary described what happened yesterday – by saying that it rained all day.

b. Mary described how she found John – by saying that he was happy.

The difference predicates of description care about

Explicit fact descriptions fully display the nature of the fact being referred to, but not so event descriptions.

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Handout 3
The Metaphysics of Relations

1. Relations and their linguistic expressions

Linguistic examples of relations:
(1) a. John likes Mary.
b. The house is next to the church.
c. The apples resembles the pear.

Which linguistic expressions express relations?
Verbs: *like, resemble, have*
Prepositions: *about, near, over, in*
Nouns: *father, daughter, neighbor, relation*
Adjectives: *familiar, neighboring, related*

<u>A linguistic issue</u>
Do verbs really denote relations?
Eventive verbs
Davidsonian event semantics:
(2) a. John met Mary in the building.
b. ∃e(meet(e, John, Mary)) & in(e, the building))
Neo-Davidsonian event semantics:
(2) c. ∃e(meet(e) & Agent(e, John) & Theme(e, Mary) & in(e, the building))
Thematic relations on Davidsonian event semantics:

Relational features of individuals

Thematic relations on Neo-Davidsonian event semantics:

Relations between individuals and events

Agent(John, e), theme(Mary, e)

This means eventive verb meaning involves two-place thematic relations and one-place event predicates.

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2. Types of relations

Two-place (dyadic) relations:

Liking, resembling, being near to

Three-place relations:

(3) a. x is between y and z.

b. John compared Mary to Bill.

c. John sent Mary a book.

Four-place relation:

(4) Paris is more distant from Rome than Rome is from Naples.

Can four-place relations be expressed by simple predicates in natural language?

Are three-place relations the limit for simple predicates?

Or event two-place relations?

(5) a. x is between the two things, y and z

b. John compared two people: Mary and Bill

c. John send [Mary (HAVE) a book]

<u>Asymmetric relations</u>

(6) a. Joe is taller than Bill.

b. Bill is taller than Joe.

(7) a. The box is inside the wardrobe.

b. ??The wardrobe is inside the box.

Symmetric relations and plural properties:

(8) a. Joe is as tall as Bill.

b. Bill is as tall as Joe.

c. Joe and Bill are equally tall.

(9) a. John resembles Mary.

b. Mary resembles John.

c. John and Mary resemble each other.

(10) a. Joe is a neighbor of Bill.

b. Bill is a neighbor of Joe.

c. Joe and Bill are neighbors.

Linguistic hypothesis

Predicates expressing symmetric relations always come with a variant expressing a plural property.

Metaphysical hypothesis

Symmetric relations are in fact plural properties (Dixon 2018).

Unigrade and multigrade relations

Unigrade relation: relation with a fixed arity (always two-place, three-place etc)

Multigrade relation: relation with variable arity

Example:

Entailment (different concusions may require different number

Multigrade relation vs plural property:

(11) a. Premise 1, premise 2, and premise 3 entail that p.

b. Premises 1, 2, and 3 entail that p.

Potential case of a multigrade relation: mathematical operations, listings

(12) a. John added two and two and two.

b. John wrote down two and three and two.

c. The catalogue lists apartment 1 and apartment 2 and again apartment 1.

Multigrade relation vs multigrade argument position :

A multigrade relation is a relation of variable adicity

Can be completed by any number of objects, in a particular order.

A multigrade argument position is an argument position of variable adicity: can be filled by a variable number of objects in a particular order.

A particular argument position can be multigrade or plural

Consequence:

Two non-Boolean meanings of *and*:

- Plurality forming
- List-forming

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3. Internal and extensional relations

Internal relations:

'is taller than' – comparatives in general, 'resembles'

External relations

is near to, 'likes', 'owns', …

(13) R is an internal relation that holds between x and y

- iff R obtains necessarily. between x and y.
- iff R obtains in virtue of the properties of x and y.

R is an external relation iff R is not an internal relation

<u>Notions of internal relation</u>

Moore:

(14) a. R is internal iff if xRy, then x bear R to y necessarily

Armstrong:

(14) b. R is internal iff if x R y, then R holds of x and y in virtue of their intrinsic (nonrelational) properties

Lewis:

(14) c. R is internal iff is x R y, then R supervenes upon the intrinsic natures of x and y

R is external iff if xRY, then R does not supervene on the natures of x and y separately, but supervenes on the nature of the composite of x and y.

Supervenience:

(15) x supervenes on y if any a change in x is necessary for a change in y to be possible.

Examples:

x is the same shape as y: internal relation (not for Moore)

is a different shape than: internal

x is taller than y: internal

In fact, all comparative relations are internal.

Also: *similar, resemble, like*

x is one meter distant from y: external relation

Questions:

- Are internal relations needed?
- Are external relations needed?

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3. Bradley's regress

Bradley Regress

External relations lead to a vicious (infinite) regress:

For an external relation to be something to objects x and y and to x relate to y, there needs to be a subsidiary relation R' relating x and y to R. But then there must be a further subsidiary relation R'' relating x and y (and R) to R', and then the same holds for R'' etc.

Bradley's conclusion

External relations can be eliminated from our ontology.

Strategy to overcome Bradley's regress

Do not make use of general relations, but of particulars:

1. Facts
2. Tropes

1. Using facts

If xRy, then the fact that xRy exists and ensures that x and y are related by R.

The problem, according McBride:

Facts presuppose general relations being able to hold between entities, they are not pluralities of the sort R, x, y.

Possible response:

General relations *are meant to* constitute facts when applying to entities.

Compare Frege's view about concepts:

Concepts are unsaturated entities: they by nature need to apply to entities to form propositions
Also a comparison can be made to the composition of structured objects:
Example of a ham sandwich, consisting of a slice of bread x and a piece of ham y
R 'being directly beneath:
X, y, R make a ham sandwich just in case xRy.
The form is partly constitutive of the whole (brings the parts together) and not an extra part
This is a particular version of hylomorphism, the Aristotelian view that objects are composed of both parts and structure (Mark Johnston 'Hylomorphism')

<u>Change of perspective: conceive of the whole as prior to the parts</u>
A form / relation is meant to make up a whole / a fact (or situation) when applied to matter or entities; it can only be understood in terms of that purpose.

<u>2. Using tropes</u>
Relations as nontransferable tropes:
A relation is essentially borne by the entities it relates.
It is in the nature of a relational trope r to be borne by particular objects x and y.
If a relational trope r exists, x and y exists, then x r y.

Thus, it in the nature of relations, as relational tropes, to relate particular entities.

Given the trope view of relations: how to make sense of relations as general relations?
Conceive of them as classes of resembling relational tropes.

<u>McBride's objection</u>
Relational tropes presuppose a solution to the problem of how two entities can be related, rather than presenting a solution to it.
'Positing relational tropes does not provide an explanation of how a relation can succeed in relating some things without requiring to be related by a further relation to them.'

But, on the relational trope view, a relation by nature succeeds in relating two things!

<u>Another solution to Bradley's regress problem (McBride)</u>

Bearing of a relation is not itself a further relation or requires a further relation.
Standing in a relation is primitive.

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4. Internal relations

Internal relations – thin relations, formal relations
Comparative relations, resemblance, identity, difference
Are internal relations dispensable?
For an internal relation to exist it suffices that te relata have a particular intrinsic character.

<u>The reductionist argument</u>
Given that internal relations are determined by the intrinsic nature of the things they relate to, they are not needed as part of our ontology.

McBride: Internal relations still play an explanatory role.
Distinguish between ontological commitment and truthmakers:
(16) Mountain Everest is taller than Mont Blanc.
Truthmaker: the height of Mount Everest and the height of Mont Blanc
Ontological commitment: internal relation between Mount Everest and Mont Blanc

Internal relation explains why the truthmakers, the heights of the two mountains, make the sentence true.
ME being taller than MB supervenes on the height of ME and the height of MB, but the internal relation between the heights explains why that is so.
Internal relations are indispensable for our theorizing about the world.

Alternative to McBride: ontological commitment to be defined in terms of truthmaking.

<u>Two notions of truthmaking</u>
Truthmaking 1 – the older notion of truthmaking:
The things in the world in virtue of which a sentence is true
Truthmaking 2 – the notion of truthmaking n truthmaker semantics:
The state / situation completely relevant for the truth of a sentence

(16) Mount Everest is taller than Mont Blanc.
Truthmaker 1: the heights of ME and MB
Truthmaker 2: the state of ME being taller than MB

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5. The problem of the order of arguments of relations

Binary non-symmetric relations appear to involve a direction of the order of arguments:
xRy does not entail yRx: the difference is a difference in direction

<u>The converse R* of a relation R</u>
(17) For a relation R, if xR*y iff yRx.

Completion of a binary relation: a relation applying to two objects in a particular order.

<u>What completions could be:</u>
States / states of affairs, facts

<u>General assumptions about non-symmetric binary relations:</u>
(18) a. <u>Identity</u>
Every non-symmetric binary relation has a distinct converse.
Examples:
over – under, taller than – smaller than,

(18) b. <u>Converses</u>
A completion of a relation is identical to the completion of its converse.
Example:
The cat being on top of the mate = the mat being underneath the cat.

(18) c. <u>Uniqueness</u>
There is a unique completion of a relation applying to two arguments (in a particular order).

Problem:
If any binary nonsymmetric relation has a distinct converse, there won't be a unique completion of a given relation applied to two arguments.

Even if Converses does not hold, there is a problem: the problem of direction:

xRy: why should R come with a direction applying first to x, then to y, or to x and y in one order; rather than with another direction

Kit Fine:
Relations are neutral: they do not come with a direction or order among their arguments.

But R applies to x differently than to y.
Explain differential application without direction:
1. Positionalism:
Relations come with positions; relations apply to entities relative to positions.
2. Antipositionalism
Relations apply to a plurality of entities in 'different ways'
3. Primitivism
The fact that an entity bears a relation to another entity cannot be explained further.

<u>Positionalism</u>
Relations come with distinct positions
Relations are completed by entities with respect to particular positions.
The problems:
1. How can positions be distinguished for symmetric relations?
There are no grounds for an entity to full in one position rather than another.
2. How should one make sense of an ontology of positions?

Potential solution to 1 (Dixon 2018):
Symmetric relations are in fact plural properties.

A problem:
An object standing in a symmetric relation to itself:

Solution: multiset, i.e., an object can occur twice in a plural argument position.

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Situation-Based Approaches in Semantics
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Handout 4
The Metaphysics of relations 2: The problem of the order of arguments and thematic relations

1. Kit Fine on neutral relations

<u>Standard assumption about relations</u>
Relations apply to entities in a particular order, that is, relations are biased or come with a direction:
'Likes' applies to a and b, in that order.

<u>Completions of relations in Fine's sense</u>
- Propositions, states of affairs
- Facts
- More generally: 'complexes'
-

Question do the differences between those things not matter?

<u>Facts and assumptions about the metaphysics of relations</u>
<u>Fact 1</u>
Every binary relation has a converse.
Every n-ary relations allows for permutations.

Linguistical reflections:

1. Passive

(1) a. John saw Mary.

b. Mary was seen by John.

2. Alternations with double-object constructions

(2) a. John gave Mary a book.

b. John gave a book to Mary.

What happens semantically with such alternations?

Change in the order of arguments?

<u>Fact 2</u>

Completions of relations and their converses / permutations are identical.

(3) a. The fact that John likes Mary = the fact that Mary is liked by John

b. The state in which the cat is on the mat. = the state in which the cat is beneath the mat.

Events?

Not mentioned by Fine!

(4) a. The event of John hitting Bill occurred this morning.

b. The event of Bill being hit by John occurred this morning

(5) a. The event of John reading the book was surprising.

b. The event of the book being read by John was surprising.

Actions?

(6) a. The act of John giving a book to Mary

b. ??? The act of Mary being given a book by John.

<u>Complex actions / dual actions</u>

Selling – buying

(7) a. John sold the car to Bill.

b. Bill bought the car from John.

<u>Two distinct actions described</u>

(8) a. Selling the car was easy; buying the car was not.

b. John planned to sell the car.

c. Bill planned to buy the car.

(9) a. John sold the car to ill with difficulty

b. Bill bought the car from John with difficulty.

Teaching – learning:

(10) a. Juan taught Mary Spanish with a lot of effort.

b. Mary learned Spanish from Juan with a lot of effort.

(11) a. ?? the action of John teaching Mary Spanish = the action of Mary learning Spanish from Juan.

b. learning Spanish is easy.

c. teaching Spanish is easy.

Consequence:

Events / actions are not just the obtaining of a relation among entities.

Dual actions:

Actions with an essential tie between two subactions.

Seeling / buying and teaching / learning describe such dual actions un a non-symmetric way.

One action is foregrounded the other backgrounded?

Agent-related adverbials apply to foregrounded subaction.

<u>Standard assumption 3:</u>

Distinct relations lead to distinct completions.

But this means that relations and their converses /permutations cannot be distinct!

<u>Fine's conclusion:</u>

Relations are unbiased or neutral as regards any direction of application or order among their arguments.

But then how are neutral relations completed by entities?

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2. Possible solutions

2.1. Positionalism

Relations are associated with a number of positions, as entities in their own right.

Assumptions about positions:

- They are distinct
- They are not ordered
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Relations and their converses / permutations involve the very same positions.

Relations apply to entities relative to particular positions.

Completions based on positions involve neutral / unbiased relations.

<u>Problems for positionalism (Fine)</u>

- The ontology of positions: positions as entities in their own right?
- Symmetric relations

<u>The ontology of positions</u>

Positions as 'entities', as 'holes', as 'places' in a derivative sense.

How plausible is positionalism linguistically?

Rather plausible, it seems:

The light noun *place*:

(12) John chose Mary in place of Sue / instead of Sue.

Transition to <u>roles</u>, which are rather plausible as entities in their own right:

(13) a. John used to run the business. Now Mary has taken his place.

b. John started organizing the conference, then Mary took his place.

The verb *replace*:

(14) John chose Mary for the job; then he replaced her with Sue.

(15) a. John had fulfilled various roles in his life.

b. Mary has to fulfill several roles at once.

<u>Roles</u>

Slots in organized events that can in principle be filled in by different individuals.

Positions:

Slots in relations that can be occupied by different individuals.

Roles, parts, and places:

Are well-reflected in natural language

Roles in events (that have been planned):

(16) a. The priest played a role in the ceremony.

b. The priest was part of the ceremony.

(17) John took part in the discussion.

The problem of symmetric relations

(18) a. John is as tall as Bill.

b. John resembles Bill.

c. John is a neighbor of Bill.

For symmetric binary relations there are no grounds for picking out one position rather than the other for completing the relation.

A potential solution (Dixon 2017):

Symmetric relations involve a single argument position: a plural argument position.

There seems good linguistic plausibility for that view:

(19) a. John and Bill are equally tall.

b. John and Bill resemble each other.

c. John and Bill are neighbors.

But what about a single entity standing in a symmetric relation to itself?

(20) a. John resembles John.

b. John is as tall as himself.

Dixon's solution:

Plural argument position are multisets:

The same entity can occur twice in a multiset.

2. Fine's suggested solutions:

2.1. Relations apply to a pluralities in a particular manner – and can apply to them in different manners

2.2. Substitution as primitive

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3. Thematic relations in linguistics

Introduction of the notion of a thematic relation in the 1960s by Jeffrey Gruber, Charles Fillmore, also Ray Jackendoff.
The question thematic relations are to answer:
How do NPs in a sentence relate to the content of the ber – the event described by the verb?

<u>Incomplete list of thematic relations</u>

Agent

Entity that deliberately performs the action (e.g., *John walked*.)

In syntax: subject of intransitive verb.

Experiencer

the entity that receives sensory or emotional input (e.g., Susan heard the song.)

Stimulus

entity that prompts sensory or emotional feeling (e.g. *David loves onions*.).

Theme / Patient

Entity that undergoes the action but does not involve change (e.g. , *I put the book on the table*)

In syntax, the theme is the direct object of a ditransitive verb.

Patient

undergoes the action and undergoes change (e.g. *The falling rocks crushed the car*.).

In syntax, the patient is the single object of a (mono)transitive verb.

Instrument

used to carry out the action (e.g. *Mary cut the ribbon with a pair of scissors*.).

Location

Region where the action occurs (e.g., *They played in the garden*)

Direction or goal

where the action is directed towards (e.g., *He walked to school.).*

Recipient

Entity acting as goal with change in ownership, possession (e.g., *Joe gave the book to Mar*y).

In syntax, the recipient or goal is the indirect object of a ditransitive verb.

Source or origin

Place where the action originated (e.g., *John walked away from the tree.)*

Time

the time at which the action occurs

Beneficiary

the entity for whose benefit the action occurs (e.g., *I baked him a cake*)

<u>Theta relations</u>
The question theta relations are meant to answer:
how are syntactic positions of NPs aligned with particular argument positions of the verb?

The distinction between thematic relations, theta-roles and grammatical functions (adopted from the Wikipedia entry on thematic relations:

<u>Thematic relations</u>

are purely semantic descriptions of the way in which the entities described by the noun phrase are functioning with respect to the meaning of the action described by the verb. A noun may bear more than one thematic relation. Almost every noun phrase bears at least one thematic relation (the exception are expletives). Thematic relations on a noun are identical in sentences that are paraphrases of one another.

<u>Theta roles</u>

are syntactic structures reflecting positions in the argument structure of the verb they are associated with. A noun may only bear one theta role. Only arguments bear theta roles. <u>Adjuncts</u> do not bear theta roles.

<u>Grammatical relations</u>

express the surface position (in languages like English) or case (in languages like Latin) that a noun phrase bears in the sentence.

Thematic relations

concern the nature of the relationship between the meaning of the verb and the meaning of the noun.

Theta roles

are about the number of arguments that a verb requires (which is a purely syntactic notion). Theta roles are syntactic relations that refers to the semantic thematic relations.

Example :

() Joe gave the book to Mary on Friday.

Thematic relations

Joe is doing the action so is the agent, but he is also the source of the book (note Joe bears two thematic relations)

The book is the entity acted upon so it is the patient.

Mary is the direction/goal or recipient of the giving.

Friday represents the time of the action.

Theta roles

The verb *give* assigns three theta roles :

Agent : Joe

Patient/ theme : *the book*

goal/recipient : *Mary*

On Friday does not receive a theta role from the verb, because it is an adjunct.

Joe bears two thematic relations : Agent and Source

but only one theta role : the argument slot associated with the thematic relations.

Grammatical relations

The subject (S) of this sentence is *Joe*, the direct object (O) is *the book*, *to Mary* is the indirect object, and *on Friday* is an adjunct.